

Leonardo G. Clarke

475 Portola Plaza • Los Angeles, CA, 90095 • leoclarke@astro.ucla.edu

Education

University of Minnesota, Twin Cities, 2021

B.S. in Physics and in Astrophysics with Honors

Summa Cum Laude

Research Mentor: Professor Claudia Scarlata

University of California, Los Angeles, (Current)

Ph.D. Candidate in Astrophysics (projected graduation: 2027)

Research Advisor: Professor Alice Shapley

Research Interests

Galaxy formation and evolution

The interstellar medium at high redshift

Star formation and chemical evolution at high redshift

Awards and Honors

NSF Graduate Research Fellowship Program, 2022 (Honorable Mention)

Competitive Edge Fellowship, 2021, (UCLA Fellowship Recipient)

Eugene Cota-Robles Award, 2021 (UCLA Fellowship Recipient)

Chambliss Astronomy Achievement Student Award, 2021 (Honorable Mention)

AUI Board of Trustees NAC Bridge Scholarship Award, 2021 (NRAO Scholarship Recipient)

Invited Talks and Presentations

Conference - Observing and Simulating Galaxy Evolution in the Era of JWST: Congressi Stefano Franscini conference centre at Monte Verità, Ascona, Switzerland, August 22, 2024

Conference - Galfresca: Carnegie Observatories, Pasadena, CA, September 19, 2024

Physics & Astronomy Colloquium (invited talk): University of California, Davis, Davis, CA, May 15, 2025

Lunch Seminar (invited talk): Carnegie Observatories, Pasadena, CA, March 14, 2025

Conference - The growth of galaxies in the Early Universe - X (invited talk): Sesto centre, Sesto, Italy, January 21, 2025

Virtual Seminar (invited talk): University of Kansas, Lawrence, KS, April 3, 2026

Conference - The Olympian Symposium: Mediterranean Village, Paralia Katerini, Greece, May 19, 2026

Conference Organizing

Sierra Conference 2023: University of California grad student conference (UCLA campus contact person), Hodgdon Meadows, Yosemite National Park, CA

Sierra Conference 2024: University of California grad student conference (Lead organizer), Mammoth Lakes, CA

Sierra Conference 2025: University of California grad student conference (Lead organizer), Hodgdon Meadows, Yosemite National Park, CA

Publications

First Author:

Clarke, L., Lam, N., Shapley, A. E., Topping, M. W., Brammer, G. B., Sanders, R. L., Reddy, N. A., & Karthikeyan, S. "Emission-line Diagnostics at $z \sim 2$: A Probe of the Ionizing Spectrum and α Enhancement Beyond Cosmic Noon." 2026, arXiv e-prints, arXiv:2601.11092.

Clarke, L., Shapley, A. E., Lam, N., Topping, M. W., Brammer, G. B., Sanders, R. L., Reddy, N. A., & Karthikeyan, S. "The Star-forming Main Sequence and Bursty Star-formation Histories at $z > 1.4$ in JADES and AURORA." 2025, arXiv e-prints, arXiv:2510.06681.

Clarke, L., Shapley, A. E., Sanders, R. L., Topping, M. W., Brammer, G. B., Bento, T., Reddy, N. A., & Kehoe, E. "The Star-forming Main Sequence in JADES and CEERS at $z > 1.4$: Investigating the Burstiness of Star Formation." 2024, ApJ, 977, 133.

Clarke, L., Shapley, A., Sanders, R. L., Topping, M. W., Jones, T., Kriek, M., Reddy, N. A., Stark, D. P., & Tang, M. "Ultra-deep Keck/MOSFIRE Spectroscopic Observations of $z \sim 2$ Galaxies: Direct Oxygen Abundances and Nebular Excitation Properties." 2023, ApJ, 957, 81.

Clarke, L., Scarlata, C., Mehta, V., Keel, W. C., Cardamone, C., Hayes, M., Adams, N., Dickinson, H., Fortson, L., Kruk, S., Lintott, C., & Simmons, B. "An Old Stellar Population or Diffuse Nebular Continuum Emission Discovered in Green Pea Galaxies." 2021, ApJL, 912, L22.

Contributing Author:

Kehoe, E., Shapley, A. E., Sanders, R. L., Reddy, N. A., Topping, M. W., Lam, N., **Clarke, L.**, Cullen, F., Ellis, R. S., Förster Schreiber, N. M., Jones, T., Khostovan, A. A., McLeod, D. J., McLure, R. J., Narayanan, D., Oesch, P., & Pahl, A. J. "The AURORA Survey: Tracing Galactic Outflows at $z \geq 2.5$ with JWST/NIRSpec Near-ultraviolet Absorption Lines." 2025, ApJ, 994, 170.

Khostovan, A. A., Sanders, R. L., Shapley, A. E., Topping, M. W., Reddy, N. A., Garcia, A. M., Berg, D. A., **Clarke, L.**, Cullen, F., Ellis, R. S., Förster Schreiber, N. M., Glazebrook, K., Jones, T., McLeod, D. J., Pahl, A. J., Pettini, M., & Torrey, P. "The AURORA Survey: The Mass -- Metallicity and Fundamental Metallicity Relations at $z \sim 2.3$ Based Purely on Direct T_e Metallicities." 2025, arXiv e-prints, arXiv:2512.16989.

Foley, J., Shapley, A., Sanders, R., Reddy, N. A., Topping, M. W., Stanton, T. M., Pettini, M., Cullen, F., Ellis, R. S., Förster Schreiber, N. M., Jones, T., Pahl, A. J., **Clarke, L.**, & Lam, N. "The AURORA Survey: Constraining Chemical Enrichment Pathways at Cosmic Noon with Argon Abundances." 2025, arXiv e-prints, arXiv:2512.10130.

Topping, M. W., Sanders, R. L., Shapley, A. E., Pahl, A. J., Reddy, N. A., Stark, D. P., Berg, D. A., **Clarke, L.**, Cullen, F., Dunlop, J. S., Ellis, R. S., Schreiber, N. M. F., Illingworth, G. D., Jones, T., Narayanan, D., Pettini, M., & Schaerer, D. “The AURORA survey: the evolution of multiphase electron densities at high redshift.” 2025, MNRAS, 541, 1707.

Sanders, R. L., Shapley, A. E., Topping, M. W., Reddy, N. A., Berg, D. A., Bouwens, R. J., Brammer, G., Carnall, A. C., Cullen, F., Davé, R., Dunlop, J. S., Ellis, R. S., Förster Schreiber, N. M., Furlanetto, S. R., Glazebrook, K., Illingworth, G. D., Jones, T., Kriek, M., McLeod, D. J., McLure, R. J., Narayanan, D., Oesch, P. A., Pahl, A. J., Pettini, M., Schaerer, D., Stark, D. P., Steidel, C. C., Tang, M., **Clarke, L.**, Donnan, C. T., & Kehoe, E. “The AURORA Survey: The Nebular Attenuation Curve of a Galaxy at $z = 4.41$ from Ultraviolet to Near-infrared Wavelengths.” 2025, ApJ, 989, 209.

Sanders, R. L., Shapley, A. E., Topping, M. W., Reddy, N. A., Berg, D. A., Khostovan, A. A., Bouwens, R. J., Brammer, G., Carnall, A. C., Cullen, F., Davé, R., Dunlop, J. S., Ellis, R. S., Förster Schreiber, N. M., Furlanetto, S. R., Glazebrook, K., Illingworth, G. D., Jones, T., Kriek, M., McLeod, D. J., McLure, R. J., Narayanan, D., Oesch, P. A., Pahl, A. J., Pettini, M., Schaerer, D., Stark, D. P., Steidel, C. C., Tang, M., **Clarke, L.**, Donnan, C. T., & Kehoe, E. “The AURORA Survey: High-Redshift Empirical Metallicity Calibrations from Electron Temperature Measurements at $z=2-10$.” 2025, arXiv e-prints, arXiv:2508.10099.

Lam, N., Shapley, A. E., Sanders, R. L., Do, T., Jones, T., Coil, A., Kriek, M., Mobasher, B., Reddy, N. A., Siana, B., & **Clarke, L.** “MOSDEF-3D: Keck/OSIRIS Maps of the Ionized ISM in ~ 25 Galaxies.” 2025, arXiv e-prints, arXiv:2506.22547.

Pahl, A., Topping, M. W., Shapley, A., Sanders, R., Reddy, N. A., **Clarke, L.**, Kehoe, E., Bento, T., & Brammer, G. “A Spectroscopic Analysis of the Ionizing Photon Production Efficiency in JADES and CEERS: Implications for the Ionizing Photon Budget.” 2025, ApJ, 981, 134.

Shapley, A. E., Sanders, R. L., Topping, M. W., Reddy, N. A., Pahl, A. J., Oesch, P. A., Berg, D. A., Bouwens, R. J., Brammer, G., Carnall, A. C., Cullen, F., Davé, R., Dunlop, J. S., Ellis, R. S., Förster Schreiber, N. M., Furlanetto, S. R., Glazebrook, K., Illingworth, G. D., Jones, T., Kriek, M., McLeod, D. J., McLure, R. J., Narayanan, D., Pettini, M., Schaerer, D., Stark, D. P., Steidel, C. C., Tang, M., **Clarke, L.**, Donnan, C. T., & Kehoe, E. “The AURORA Survey: An Extraordinarily Mature, Star-forming Galaxy at $z \sim 7$.” 2025, ApJ, 981, 167.

Shapley, A. E., Sanders, R. L., Topping, M. W., Reddy, N. A., Berg, D. A., Bouwens, R. J., Brammer, G., Carnall, A. C., Cullen, F., Davé, R., Dunlop, J. S., Ellis, R. S., Förster Schreiber, N. M., Furlanetto, S. R., Glazebrook, K., Illingworth, G. D., Jones, T., Kriek, M., McLeod, D. J., McLure, R. J., Narayanan, D., Oesch, P., Pahl, A. J., Pettini, M., Schaerer, D., Stark, D. P., Steidel, C. C., Tang, M., **Clarke, L.**, Donnan, C. T., & Kehoe, E. “The AURORA Survey: A New Era of Emission-line Diagrams with JWST/NIRSpec.” 2025, ApJ, 980, 242.

Hviding, R. E., Momcheva, I. G., & **Clarke, L.** “Improved Empirical Backgrounds for JWST NIRISS Image/Wide-field Slitless Spectroscopy Data Reduction.” 2024, AJ, 168, 220.

Lin, Y.-H., Scarlata, C., Mehta, V., Skillman, E., Hayes, M., McQuinn, K. B. W., Fortson, L., Chworowsky, K., & **Clarke, L.** “Low-metallicity Galaxies from the Dark Energy Survey.” 2023, ApJ, 951, 138.

Sanders, R. L., Shapley, A. E., **Clarke, L.**, Topping, M. W., Reddy, N. A., Kriek, M., Jones, T., Stark, D. P., & Tang, M. “A Preview of JWST Metallicity Studies at Cosmic Noon: The First Detection of Auroral [O II] Emission at High Redshift.” 2023, ApJ, 943, 75.